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Case Study: Annular Isolation Using A Hydro-Mechanical Expansion Plug & Perforator

DayIn Beazer, Stuart Butcher, and Scott Ellery, Lee Energy Systems Inc., Calgary, Alberta, Canada; Keith Brand, Amplitude Energy, Perth, WA, Australia; MHD Tareq Alsaman, SLB, Perth, WA, Australia

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Abstract

This paper details the design, installation, and verification of the upper abandonment cement plug on a seven well abandonment campaign in the Bass Strait. A final abandonment plug was required across the 9-5/8" casing and 9-5/8" × 13-3/8" casing annulus as the final pressure-containing barrier. The paper discusses evaluating verification methods, while addressing concerns from the Australian regulator regarding cement slumping in the annulus and its potential impact on long-term barrier integrity in the well abandonment strategy.

To prevent cement slumping within the 9-5/8" casing, a bridge plug is typically used; however, no equivalent solution exists for the casing annulus. Traditional methods, such as squeezing viscous pills into the annulus, face limitations due to weak loss zones, cement displacement complications, and verification difficulties. A mechanical solution was explored by means of a hydro-mechanical expansion packer designed to expand the 9-5/8" casing against the 13-3/8" casing ID, forming a base for the annular cement squeeze.

The hydro-mechanical expansion packer (MPX) and associated mechanical barrier strategy were successfully deployed during abandonment operations on the Basker, Manta, and Gummy well campaign. Following preparatory operations, the 838 MPX was run and set at approximately 3215 ft (980 m), forming a mechanical support barrier for the annular cement squeeze. A hydro-mechanical perforating tool was used to perforate the 9-5/8" casing immediately above the MPX dual annular plug and just below the wellhead to provide access to the 9-5/8" × 13-3/8" annulus. This enabled annular circulation and conditioning prior to the cement squeeze. Flow checks confirmed the well was static and free of hydrocarbons. A 325 ft (100 m) cement plug was then placed into the annulus and casing bore above the dual annulus plug, and pressure tested to 1250 psi for 15 minutes, with a 4.5 t set-down to tag the cement plug confirming placement integrity. All cement plugs were tagged within the required depth confirming the MPX expansion and isolation of the annulus with no cement slumping. The MPX approach mitigated cement slumping risk, enabled effective annular support, and reduced the need for high-pressure testing across sensitive casing.

This paper presents a novel mechanical solution to prevent cement slumping in casing annuli during well abandonment – a long-standing industry challenge. The deployment of the hydro-mechanical expansion

packer to expand casing and form a support barrier in the outer annulus represents an innovative solution in offshore abandonments. It offers an alternative to traditional viscous pill methods, improving barrier integrity, regulatory compliance, and operational efficiency in complex well environments.

Introduction

In 2024, Cooper Energy, now Amplitude Energy (AEL) executed the full plug and abandonment (P&A) of seven subsea wells located in the Basker, Manta, and Gummy (BMG) fields in the Gippsland Basin, approximately 31 miles (50 kilometers) off the Victorian coastline in Australia. The campaign was undertaken in compliance with Cooper Energy's Well Engineering Standards, the BMG Well Operations Management Plan, and under the regulatory oversight of Australia's National Offshore Petroleum Safety and Environmental Management Authority (NOPSEMA).

The decommissioning campaign marked the first use of the HELIX Q7000 vessel in Australian waters as a dynamically positioned heavy intervention vessel configured for complex subsea well abandonment. This paper focuses on one of the technical challenges encountered during the project: establishing long-term isolation of the annular space between the 9-5/8" and 13-3/8" casing strings using a novel hydro-mechanical expansion packer (MPX) and a non-explosive hydraulic perforating tool (Gator).

Annular isolation was critical in ensuring the permanent abandonment of wells where traditional options, such as squeezing viscous pills or pulling casing, were either unfeasible or carried high operational risk (Simmadorai et al., 2021). Concerns regarding cement slumping prompted the development and field validation of a mechanical barrier strategy to support and verify annular cement plugs. This case study details the engineering, deployment, and verification of this mechanical system, emphasizing its implications for regulatory compliance, operational efficiency, and long-term well integrity.

Background and Project Context

The Basker Manta Gummy Field consisted of seven offshore subsea oil wells in the Gippsland Basin. The wells were drilled between 2006 and 2008, and the field ceased oil production in 2011 with the departure of the FPSO. Prior to the FPSO departure, the field was prepared for a Non-Production Phase (NPP) by flushing the seabed infrastructure and shutting in wells at the tree valves. Six wells remained shut-in via the tree valves, and one well had the lower wellbore suspended with cement plugs. All seven wells remained interconnected through pipework and the Basker-A manifold.

In 2024, Amplitude Energy Limited (formerly Cooper Energy) successfully completed the plugging and abandonment (P&A) of all seven wells using the Q7000 heavy well intervention vessel and novel methodologies and technology.

The well abandonments were conducted in two phases to meet the plug and abandonment objectives. The first phase is the isolation and abandonment of the reservoir via tubing plug and abandonment methods. The second phase, the installation of shallow cement plug barriers across the 9-5/8" and 13-3/8" casings to isolate shallow formations and aquifers without cutting and pulling the 9-5/8" casing.

Challenge

Decommissioning projects all come with their own set of challenges; the ones pertaining to this project were unique, including specific technical requirements, wells of especially complex design, and an offshore environment exposed to strong winds and large swell. One of the key challenges was the shallow isolation barrier across 9-5/8" $\times$ 13-3/8" casing. The annulus cement isolation and prevention of cement slumping in the annulus because of a porous formation top.

Cement within the 9 5/8" casing can be retained and prevented from slumping by means of a bridge plug; however, it is not possible to install a bridge plug within the casing annulus and cement is known to

slump within porous formations reducing the effectiveness and seal of the cement plug (Eoff et al. 2015). Traditionally viscous pills are squeezed into annuli below intended cement plug depths; however, concerns around significant open hole (weak loss zones), displacement of the viscous pill by the cement and potential for slump (i.e. barrier not in place) coupled with challenges in barrier verification (high shoe LOT in excess of 13-3/8" casing rating) have led to a need for a mechanical solution to isolate and support the annular cement placement, guaranteeing cement placement (Bagal et al. 2016).

The planning team were concerned about cement slumping in the well barriers and how this needed to be considered as part of well abandonment planning. The regulator had previously challenged the installation of shallow abandonment plugs across the 9-5/8" × 13-3/8" casings without recovering the 9-5/8" casing and expressed that Hi-viscous pills as a cement base are not a guarantee that the annulus cement is where it is intended to be.

Conditions that may result in cement slumping:

- Weak formation with a loss zone, porous isolation issues
- Drilling washouts
- Low fracture gradients
- Differing well fluid densities between the cement and the fluid below the intended plug depth
- Cement contamination, poor well cleaning

A mechanical plug or cement base would eliminate these potential conditions. AEL approached Lee Energy Systems with the challenge of using their high expansion packers as a means of expanding 9-5/8" casing to create a mechanical bridge to prevent cement slumping in the annulus. The requirement was for a packer system strong enough to expand 9-5/8" casing isolating or forming a baffle within the 9-5/8" × 13-3/8" annulus to provide a mechanical plug within the 9-5/8" casing that could be used as a cement base.

Outline of the Hydro-Mechanical High Expansion Packer Technology

A method for expanding casing to create a dual A & B annular cement barrier (isolation)

The MPX is a hydro-mechanical, high-expansion packer system (see Figure 1), designed to enable dual annular isolation in a single deployment. It features a hydraulic power section that applies axial force to the expansion element. This axial load is mechanically converted into radial expansion, driving the inner casing outward into the B annulus. The resulting radial interference creates a mechanical barrier across the B annulus, allowing fluid, such as cement or an alternative sealing material, to be effectively placed without significant loss of circulation.

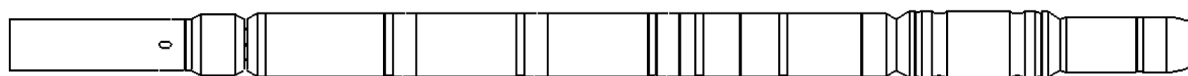


Figure 1—Hydro-Mechanical Expansion Packer

Following expansion, an internal locking mechanism secures the packer in its set position. The setting tool is then decoupled and remains in the wellbore, maintaining isolation across the A annulus (see figure 2). This configuration enables simultaneous isolation of both the A and B annuli, providing a foundation for subsequent placement of sealing materials above the packer while minimizing the risk of fluid migration or losses.

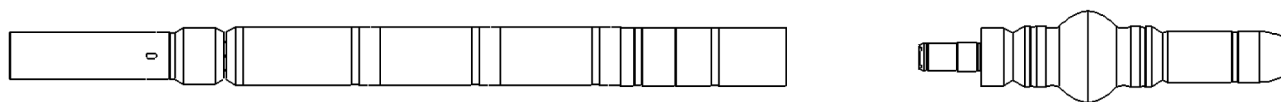


Figure 2—Decoupled Running Tool from Permanent A & B Annular Plug

The hydro-mechanical expansion packer, when activated in the well, will expand the primary casing into the formation or secondary casing string creating a barrier or restriction to reduce cement slumping and hold up the cement plug above it (see [figure 3](#)). This would solve the conditions that lead to cement slumping.



Figure 3—Hydro-Mechanical Expansion Packer Expanding Primary Casing into Secondary Casing

A test program was conducted to evaluate the performance of the 838-MPX or 8.38" OD hydro-mechanical expansion packer in two key casing configurations prior to being run on the job. The primary objective was to assess the ability of the tool to expand 9-5/8", 47 lb/ft, L80 casing beyond 12.347" OD. This would indicate expansion sufficient to simulate interference with the ID of a 13-3/8", 72 lb/ft outer casing string without compromising casing integrity. A secondary but equally important objective was to evaluate the expansion of the 9-5/8" casing while nested inside a 13-3/8", 72 lb/ft, L80 casing to confirm the tool's ability to create a fluid-restrictive barrier within the annular space. A hydraulic pressure test would test the sealing capability of the A annular plug while a hydraulic fluid test would be used to confirm the restrictiveness of the B annular plug.

Test 1: Standalone 9-5/8", 47#, L80 Casing Expansion

Assess the ability of the MPX system to radially expand 9-5/8", 47 lb/ft, L80 casing to achieve interference with the ID of a 13-3/8", 72 lb/ft outer string. The nominal ID of the 13-3/8", 72 lb/ft casing is 12.347".

The running tool and plug were assembled and placed into the 9-5/8" casing target. The tool was activated and an expansion in the casing was observed to 12.406" (see [figure 4](#)). This indicated that the casing would be sufficient to create interference with the 13-3/8", 72 lbs/ft casing.



Figure 4—Expansion of 9-5/8", 47 lb/ft, L80 Casing in Open Air to an OD of 12.406"

Test 2: Tested Casing Expansion (9-5/8" inside 13-3/8")

Demonstrate the MPX's capability to expand the inner 9-5/8" casing sufficiently to deform the surrounding 13-3/8", 72 lb/ft, L80 casing. The test aims to validate effective annular restriction and evaluate fluid seal performance between the casing strings.

The running tool and plug were assembled and placed in the test target of 9-5/8", 47 lbs/ft, L80 casing nested inside the 13-3/8", 72 lbs/ft casing. The tool was activated to achieve max expansion inside the casing target.

The fluid seal test was conducted three times to evaluate the ability of the expanded 9-5/8" casing to restrict water flow through the B-annulus. In each test, the MPX packer was fully set with the ratchet ring engaged and then released from the mandrel in 3 separate casing targets (see [figure 5](#)).



Figure 5—Hydro-Mechanical Expansion Packer in Nested Casing Target

The first test successfully held a full column of water in the B-annulus, allowing overflow at the top, with only a small drip observed at the bottom (see [figure 6](#)). This indicated a very successful expansion and great restriction in the B annulus.



Figure 6—Test 1, Small Drip

The second test also restricted flow, though to a lesser extent, with the water level stabilizing approximately 12 inches below the top, indicating great restriction with minor leakage (see [figure 7](#)).



Figure 7—Test 2, Minor Leak

In Test 3, while the 9-5/8" casing did expand, the deformation was asymmetric, resulting in a smaller outer diameter and reduced sealing effectiveness, particularly against low-viscosity water. Flow rate in = flow rate out.

Overall, the fluid seal testing confirmed that a substantial restriction was formed in the annular space, demonstrating the system's ability to limit flow across the expanded zone.

It is expected that under actual well conditions, where the B-annulus may contain fluid, mud cake, barite, or cement residue, the expanded casing is expected to conform to these materials, creating a more robust barrier and restriction with less expansion required than free pipe. This suggests a strong potential for effective isolation during cementing operations above the expanded section.

Outline of the Hydro-Mechanical Perforator Technology

Another method for perforating casing or tubing

The Gator is a hydro-mechanical perforator see [figure 8](#). It can be broken down into three sub sections or assemblies: the Cutter Sub, Hydraulic Cylinders, and the Bottom Sub. To activate the tool, the operator must apply pressure down the tubing or drill string. The pressure in the tubing string triggers the Hydraulic Cylinders which in turn push the four blades in the Cutter Sub upwards and out into the casing or tubing. This creates four holes in the casing at 90° from each other. To retract, the operator must release any stored pressure in the string and pull upwards. This upward motion forces the blades downward and resets the Hydraulic Cylinders allowing for multiple perforations to be performed in a single run.

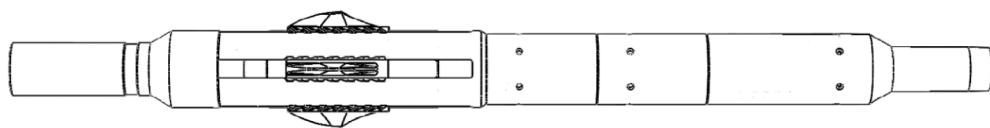


Figure 8—Hydro-Mechanical Perforator

Mechanics of the Perforation

The perforating action starts when the blade engages the casing at a point initiation. As the force increases on the four points, the casing yields and the blades are allowed to penetrate through the casing in a shearing action. As the blades shear at the four shear planes 90° from each other, the top of the blades extends through and push one side of the casing outwards in a spiraling fashion, as shown in [figure 9 & 10](#).

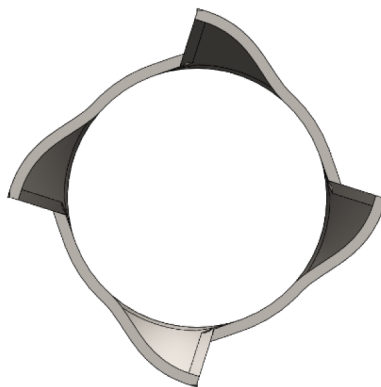


Figure 9—Model of Casing Cut Profile



Figure 10—Casing Cut Profile

As shown, the blades don't penetrate directly through into the casing, rather they create a cut and push the casing outwards in the localized cut areas, elongating the growth of the perforations and increasing the total flow area. The mechanical perforating action provides access windows to the outer annulus.

Because of the unique perforating mechanics and the controlled depth of cut, the blades will not damage outer casing strings or conductor pipe. Other benefits include minimal casing deformation allowing full drift of casing post activation; as well as eliminating perf burs or debris left inside the casing post activation which permits high tolerance tools such as packers or cup tools to pass through unscathed.

Cut holes vary in size depending on the casing weight and size of tool. An example of the holes created in 9-5/8" casing from the 825 Gator compared with the holes from a conventional explosive perforation of a 39 Gram BH charge are as follows:

Conventional Perforation specification of a 39 Gram charge:

- Total flow area per charge is .88 in²
- Total flow area for an 18 shots/ft gun over 3 ft is 47.52 in²

825 Gator Perforator Specs

- Total flow area per hole is 5.38 in²
- Total flow area per cut (4 holes) is 21.50 in²

Total flow area of 1 activation/ft (16 holes) over 3 ft is 64.50 in².

Testing the Mechanical Perforator in Eccentric Uncemented Casing

In order to ensure outer casing integrity, a workshop test was performed that employed a 9-5/8" casing welded inside a 13-3/8" casing joint. The casing joint was welded in an eccentric manner with the 9-5/8" casing joint laying on the low side of the 13-3/8" casing as seen in [figure 11](#) left. The mechanical perforator was employed to cut the 9-5/8" casing. Once the tool was successfully activated, the 9.5/8" casing joint was cut in half to see the 9-5/8" lifted off the 13-3/8 as shown in [figure 11](#) right.



Figure 11—Welded Test Target, left; Internal CSG Lifted from External CSG Post Perforation, right

The test successfully confirmed that the outer joint was undamaged. It also established that the mechanics of the cut effectively lift the inner casing string off from the low side and helps to centralize it for better cement remediation or more effective clean out of the casing annulus.

When the test target was taken apart for inspection, the perforation sizes around the selected casing joint varied slightly as seen below in [figure 5](#). The mechanical perforations were much larger on the high side than on the low side. On the low side, the mechanical perforator did penetrate the casing wall and did not create as large of a window. Instead, it protruded the 9-5/8" casing joint and lifted it off from the low side of the 13-3/8" casing (see [figure 12](#) left). There was no internal scoring or marks from the mechanical perforator blade on the 13-3/8" casing (see [figure 12](#) right). Two test cuts were made on this target. One was performed in the middle of the test target and one close to the ends where the welds held the two casing strings together. In the case where the perforations were completed close to the welded ends, a small bulge was noticeable only by running a hand on the inside or outside of the outer casing string. The test perforation made in the middle of the test target left no indication of deformation on the 13-3/8" casing string.



Figure 12—Low Side Perforation, left; High Side Perforation, middle; 13-3/8" CSG Post Perforation, right

Operational Methodology/Strategy

The operational strategy will involve deploying a hydro-mechanical packer expander to expand the casing in the well. This expansion will prevent cement losses by creating an A and B annular barrier capable of supporting the cement plug. Once this step is complete, a hydro-mechanical perforating tool will be run to create perforations in the casing, enabling circulation of the annulus. The annulus will then be conditioned, and a cement plug will be placed across both the A and B annuli to complete the abandonment barrier.

To establish an A and B annular barrier, the 838-MPX will be deployed to expand the 9-5/8" casing and reduce the cross-sectional area of the B annulus inside the 13-3/8" casing. The expansion will be carried out over a 1-m interval in the wellbore. Once maximum expansion is achieved and the element is fully set against the 9-5/8" casing, the plug will be released from the setting tool by rotating the string, leaving a permanent plug in the well. This plug will provide a mechanical barrier capable of supporting cement during subsequent abandonment operations. The reduced annular volume in the B-annulus will act as a restriction and mechanical blockade to retain viscous fluids and ensure stable placement of the cement barrier above the expanded section.

Following expansion, the 825 Gator perforator BHA will be run in combination with a packer. The mechanical perforations will allow circulation of the B-annulus to remove debris and condition the annular space ahead of cement placement. Once circulation is confirmed, and the well is conditioned, cement will be

pumped across the perforations and on top of the mechanical plug. This sequence will ensure the installation of a continuous, well-supported cement barrier across both annuli.

Run #1 Setup

The casing expansion BHA consists of a handling pup joint, the 838 MPX Setting Tool, and an 838 MPX Plug assembly. With the plug located at the setting depth, the workstring/BHA will be pressured up to 3,000 psi. This will provide sufficient setting force to create an A-annulus seal and expand the 9-5/8" casing to the 13-3/8" casing to provide a base for annular cement. The plug will be released from the BHA with 18 right hand rotations, and the BHA can be pulled out of the well. See [figure 13](#) below for the BHA schematic.

LEE ENERGY SYSTEMS		Mega Packer BHA Assy 9-5/8" Csg				R&D SOLUTIONS	
		Cooper Energy Basker 4		Date: 2023-05-05			
#	Tool Layout	Mega Packer Assembly Primary	Qty	ID (in)	OD (in)	Length (ft)	Total BHA (ft)
IMPERIAL				(mm)	(mm)	(m)	(m)
1		Top of Pup to Centre Element = 10.78m					
2		Handling Pup - 10ft IOT Rental PCS-PJ-439	1	2.812	6.525	10.00	10.00
3		4-1/2" IF (NC-50) Box x Pin Connection		71.42	165.74	3.04	3.04
4							
5		Mega Packer Body 838-MPX-002	2	1.630	8.380	24.27	24.27
6		4 1/2" IF (NC-50) Box Connection		41.40	212.85	7.50	10.54
7		Fish Neck on MPX 0.59m OD = 167mm					
8							
9		Packer Assembly 838-MPX-A36-004	3	1.630	8.380	3.51	3.51
		3-1/2" IF (NC-50) Box		41.40	212.85	0.99	11.53
		Top To Mid Element 10.78m					
		Bottom To Mid Element 0.65m					

Figure 13—Hydro-Mechanical Expansion Packer BHA Run 1

BHA Run 1 Component Description

Handling Pup

Provide a location for the rig to pick up and maneuver the BHA from horizontally on the rig floor and place vertically in the well and slips.

MPX Setting Tool

The MPX Setting Tool is the power section that will translate the differential pressure applied to the workstring/BHA into mechanical setting force into the plug section. The max pressure for the tool is 4,500 psi.

MPX Plug

The MPX Plug consists of a rubber sealing/expansion element, check valve, and internal ratchet ring. The force from the MPX Setting Tool stroking will compress the element to provide a seal against the 9-5/8" casing ID, at higher pressures this will begin to expand the 9-5/8" casing until it meets the 13 3/8"s casing ID. As the plug begins to stroke/expand, the ratchet ring stops the plug from relaxing and locks in the setting force.

Run #2 Setup

The perforating and circulating BHA consisting of a handling pup, AS1-X Packer (mechanical set packer), ball drain sub, unloader sub, 825 Gator, and velocity control valve (VCV). The Gator blades will be positioned at the deepest planned perforation/zone and pumping will begin down the workstring to close the VCV. Once the VCV has closed, the pressure in the workstring/BHA will be increased to 4,000 psi. Pressure will then be bled off at surface, which will in turn open the VCV valve. To retract the blades, the workstring will be moved up hole and the blades can be positioned at the next perforation depth. The perforating procedure will be repeated across the 2 intervals of interest. To circulate the B-annulus, the AS1-X packer will be set between the 2 perforation intervals. Pumping will begin down the workstring, out the VCV, into the lower interval, up the B-annulus, and back into the A-annulus via the upper interval. Once the clean out is complete, the packer can be unset, and the BHA can be pulled out of the well. The tubing Ball Drain sub can be opened for higher circulation rates or cementing operations. See below [figure 14](#) for the BHA 2 schematic.

LEE ENERGY SYSTEMS		PCK/BDS/ULS/HWG/VCV BHA Assy 9-5/8" Csg				R&D SOLUTIONS	
		Cooper Energy Gippsland Basin		Date: 2023-05-03			
#	Tool Layout	825 Gator BHA Basker 4	Qty	ID (in)	OD (in)	Length (ft)	Total BHA (ft)
IMPERIAL Top of Pup to Blade Center = 8.41m							
1	1	Pup Joint 4-1/2" IF (NC50) x 6ft OM15372	1	2.812	6.625	98.81	98.81
		4-1/2" IF (NC-50) Box x 4 1/2" EUE Pin Connection		71.42	8.20	2.45	2.45
2	2	9 5/8" ASX-1 Packer 02	1	2.750	6.250	98.03	196.84
		4-1/2" EUE Box x Pin Connection(Cooper Owned)		3.50	8.20	2.42	4.87
3	3	Crossover- OM15374	1	2.82	6.900	31.49	228.33
		4-1/2" EUE Box x 4 1/2" IF (NC-50) Pin Connection		71.50	175.26	0.69	5.56
4	4	613-BDS-004 Ball Drain Sub	1	1.000	6.500	27.16	255.49
		4-1/2" IF (NC-50) Box x Pin Connection		25.40	165.10	0.57	6.12
5	5	650-TW-ULS-012 Unloader Sub	1	2.000	6.500	50.39	305.88
		4-1/2" IF (NC-50) Box x Pin		50.80	165.10	1.32	7.44
6	6	825-HWG-003 Series Gator Perforator	1	1.250	8.250	183.07	488.95
		4-1/2" IF (NC-50) Box x Pin Connection		31.75	209.55	4.65	12.09
7	7	825 Gator Perforator Pistons x 8	1	2.563	6.250	0.00	488.95
		4-1/2" ACME Pin x 4 1/2" IF (NC-50) Pin Connection		65.09	158.75	0.00	12.09
8	8	Crossover OM15376	1	2.250	6.900	31.50	520.45
		4 1/2" IF Box x 3 1/2" IF Pin (~180 l/min to activate closed)		57.15	175.26	0.70	12.79
9	9	VCV-475-018 Velocity Control Valve	1	0.500	6.250	48.81	569.26
		3-1/2" IF (NC-38) Box		12.70	158.75	1.23	14.02
Top of BHA to packer mid element 3.57m							
Top of BHA to mid Gator blade 8.41m							
Bottom of BHA to Gator mid blade 5.59m							

Figure 14—Hydro-Mechanical Perforating Tool, BHA Run 2

AS1-X Packer

A mechanically set retrievable packer used for well testing. Can be set in compression or tension and has an internal bypass for equalization. ¼ turn right hand to set and ¼ right hand to release.

Ball Drain Sub

A ball drop activated circulation sub. When no ball is present, this tool allows internal flow and is sealed to the annular. To activate a ball is dropped down the workstring and once seated in the tool, pressure is applied to shift the internal sleeve and allow large TFA flow to the OD of the BHA, no internal flow will go down past the tool.

Unloader Sub

Push/Pull equalization valve. Moving string into compression opens the valve and equalizes tubing to the annulus. When the string is in tension, the valve is closed. The ULS has approx. 6 in functioning stroke.

Gator Perforator

Hydro-mechanical perforating system that uses 4 blades to perforate casing. A pressure fluctuation can be seen on surface, indicating the Gator blades have penetrated the casing to create perforations.

Velocity Control Valve

A rate-controlled valve which remains open to flow through the bottom until a predetermined rate is reached to close the valve. When the valve is closed fluid flow will stop and allow a controlled increase in workstring pressure. When pressure is bled off of the closed valve, the VCV will open and flow through the tool will continue.

Operational Execution

An 838-MPX was made up and deployed on DS55 drill pipe. The tool was run in hole and successfully set at 3194 ft (973.5 m) MD. During setting, a pressure event was recorded at 2,300 psi, indicating initiation of plug expansion. The running tool was released from the packer with 18 right-hand turns, after which a 15-minute flow check was conducted. The BHA was then pulled out of hole.

Following this, an 825 Gator tool was deployed to hydro-mechanically perforate the 9-5/8" casing to provide annular access for circulation, confirm the absence of hydrocarbons, and prepare for aquifer barrier installation. The 825 Gator perforated the casing at 3156 ft (962 m) MD, and the well flow-checked static, confirming no hydrocarbon flow. Additional perforations were made at 3152 ft (961 m) and 3150 ft (960 m), followed by perforations at 590 ft (180 m), 587 ft (179 m), and 584 ft (178 m). The well remained static throughout. The Gator tool was then run back in hole, and an ASI-X packer was set at 3090 ft (942 m) MD. The ROAM annular BOP was closed, and circulation was established through the B-annulus using the cement unit, with returns directed to the solids handling package. A total of 250 bbl of seawater was circulated, confirming the B-annulus was clean. A 15-minute flow check again verified the well remained static.

The annular BOP was then opened, the packer was unset, and a 1.5" ball was dropped to shear out the Gator circulation sub. The Gator was positioned with its blades at 3150 ft (960 m) MD, and the circulation sub was sheared open in preparation for cementing.

The ASI-X packer was then set at 3123 ft (952 m) MD, and the cementing assembly with a side-entry sub was made up. Cementing operations began with a 5-bbl mud push spacer, followed by 46 bbl of 15.8 ppg Class G cement, and a 5-bbl mud push spacer. The cement was squeezed into the 9-5/8" × 13-3/8" B-annulus. After pumping, the ASI-X packer was released, and displacement continued with 12.9 bbl of seawater while pulling out of hole, placing a balanced 328 ft (100-m) cement plug from 3192 ft (973 m) to 2864 ft (873 m) MD. The Gator was pulled out of hole, and the tools were laid out.

To confirm plug integrity, an 8-1/2" mill bit was made up and run in hole to tag the top of cement at 2877 ft (877 m) MD. The string was then picked up to 2847 ft (868 m) MD, the ROAM annular BOP was closed, and the aquifer plug was successfully pressure tested to 1,250 psi for 15 minutes (see [figure 15](#), right). With

the test complete, the annular was opened, the BHA was pulled out of hole, and the ROAM was prepared to move to the next well. See [figure 16](#) for the wellbore schematic.

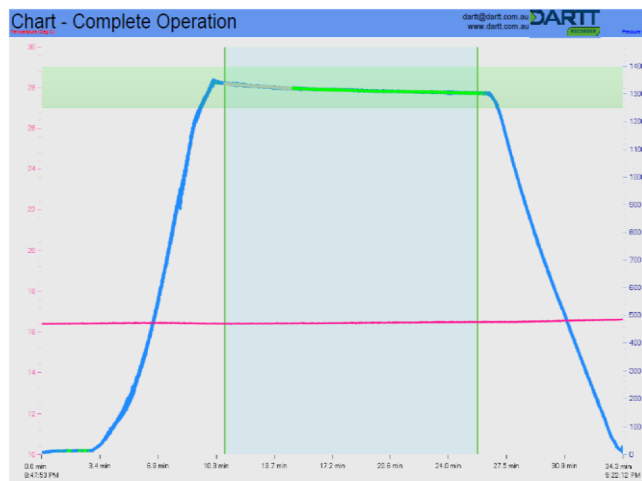


Figure 15—Pressure Test

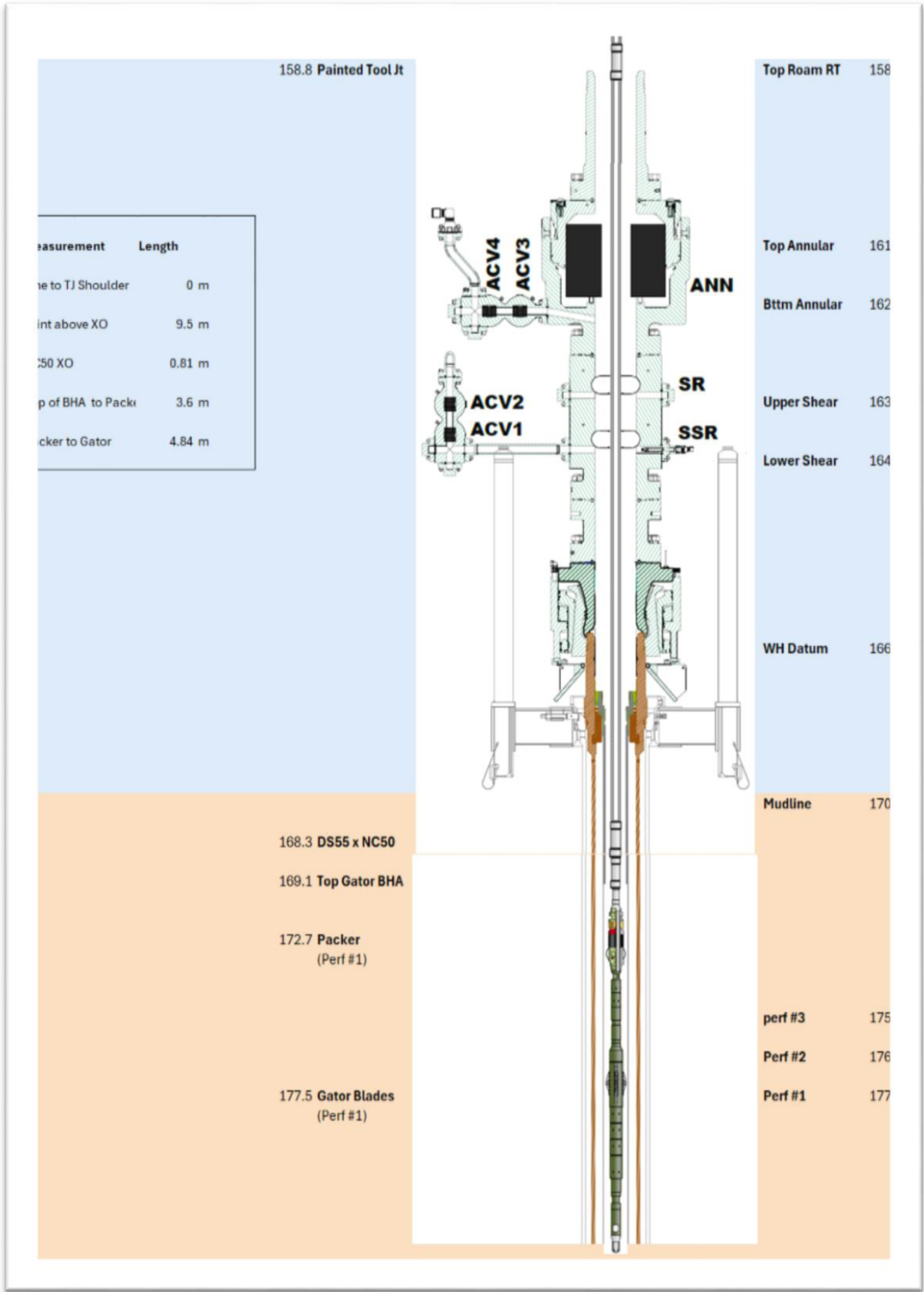


Figure 16—Wellbore Schematic

Conventional Annulus Slurry Squeeze Challenges / Limitations

In conventional plug and abandonment (P&A) operations, the standard approach to establishing a permanent barrier involves perforating the casing and pumping cement into the annular space between the casing and

the formation (Simmadorai et al., 2021). This process is intended to restore or establish zonal isolation by filling the annulus with cement that can set and form a hydraulic seal. However, despite its widespread use, this method presents significant operational and integrity-related challenges, particularly in ensuring that the cement remains in place and forms a competent seal throughout the target interval. One of the primary concerns with this approach is the lack of mechanical support in the annulus after cement placement (Bagal et al., 2016). Once the cement is displaced into the annular space, it is left unsupported by any internal structure, relying solely on its own rheology and static friction against the casing and formation to maintain its position during setting. In vertical or deviated wells, this can result in gravitational slumping, where the unset cement gradually moves downward, leading to thinning of the cement sheath in the upper portion of the annulus and potential accumulation or over-thickening at the lower end. Such uneven distribution can compromise the intended barrier length and effectiveness, leaving uncemented or poorly cemented intervals that are prone to fluid migration or pressure communication between zones.

Moreover, in wells where the annulus is enlarged due to washouts or where casing centralization is suboptimal, the cement placement becomes even more unpredictable. Eccentric annuli can exacerbate displacement inefficiencies, allowing drilling mud or other fluids to remain trapped in the narrow sections while cement preferentially flows into the larger voids, further reducing the quality of the cement bond. In addition, fluid losses to the formation during the placement process can result in incomplete cement fills and reduced barrier integrity. Once the cement is placed, there is typically no way to physically contain or support it during setting, especially in long intervals, which increases the risk of channeling, micro-annuli formation, or even complete loss of zonal isolation (Joneja et al. 2016). These limitations make it difficult to consistently achieve the barrier length and integrity required by regulatory standards such as those established by API or local governing bodies.

In response to these challenges, the industry has explored various improvements, including the use of more advanced cement systems, improved displacement techniques, and enhanced zonal isolation verification methods. However, a fundamental limitation remains: the lack of structural support for the cement in the annulus during and after placement. This creates a compelling need for innovative solutions that can provide such support, especially in complex well geometries or when performing abandonment operations in aged wells with unknown or compromised annular conditions. Tools designed to offer mechanical support to the cement column in the annulus, particularly after perforation and prior to or during squeeze cementing, present a promising advancement. By addressing the root cause of cement slumping and providing a physical structure to retain cement in place, such tools can significantly improve the reliability and longevity of P&A barriers, reduce the risk of re-entry or remedial work, and enhance compliance with stringent abandonment regulations.

Slurry Properties

For this application, a custom cement slurry was specifically designed to address the challenges associated with placing cement in the annulus through perforations during plug and abandonment operations, particularly the risk of cement slumping due to the lack of mechanical support "as a contingency". The formulation was engineered to develop a high static gel strength (SGS) rapidly, thereby minimizing the time the slurry remains in a vulnerable, flowable state and reducing the potential for gravitational settling or channeling. In addition, low fluid loss properties were incorporated into the design to limit dehydration and ensure full-volume retention within the annulus. The slurry's rheological profile, specifically its plastic viscosity and yield point, was optimized to maintain adequate suspension of solids during placement and promote uniform distribution in the presence of irregular or eccentric annuli.

This slurry design allows for improved stability and placement control, reducing the likelihood of cement fallback and enhancing overall barrier integrity. The chosen properties also support effective displacement through narrow perforation tunnels and across uneven annular geometries, which are common in aged wells.

By tailoring the cement system to the specific operational risks of this P&A job, and in conjunction with a tool designed to provide mechanical support to the cement column during setting, this approach significantly enhances the reliability and long-term performance of the abandonment barrier.

Cement Job Execution

The cement job was executed to install a 100-meter balanced cement plug as part of the well's permanent abandonment program. The plug was designed to span both the 9-5/8" × 13-3/8" annulus and the internal 9-5/8" casing, with the placement achieved via squeeze cementing through perforations. A total of 46 bbls of Class G cement slurry, engineered at 15.8 ppg with high static gel strength (SGS), low fluid loss, and optimized rheological properties, was batch mixed and pumped in line with the job objectives.

To ensure proper cement placement and address the challenge of cement slumping, a mechanical packer was first deployed and expanded to act as a baffle between the casing and outer annulus, thereby supporting cement placement in the 9-5/8" × 13-3/8" annular section. The 825 Gator BHA was then used to perforate the casing at both high and low intervals, creating communication paths for cement placement. A retrievable packer was set above the lower perforation to allow controlled circulation and ensure the annulus was flushed clean before cementing. Spacer fluids were employed to prevent contamination and separate fluids effectively.

The cementing sequence included a spacer-cement-spacer approach, with precise volumes and rates monitored throughout. The first 22 bbls of slurry were squeezed into the annulus to achieve 100 meters of coverage, followed by placement of the remaining 24 bbls inside the casing to form a 328 ft (100-m) balanced plug. Controlled displacement and flowback monitoring ensured accurate placement. Upon completing the squeeze into the annulus, the packer was released to allow displacement of the remaining cement into the casing. Final displacement used 12.9 bbls of seawater at 2 bpm, accounting for the calculated under-displacement and spacer volumes.

Post-placement, a cautious pull-out-of-hole (POOH) was performed to prevent disturbing the plug, with staged circulation conducted to clean the workstring and minimize the risk of contamination. The operation concluded with a waiting-on-cement (WOC) period to 500 psi as per UCA testing prior to tagging the plug.

Results & Verification

The deployment of the 838-MPX hydro-mechanical expansion packer successfully provided a solid mechanical base for cement placement across the 9-5/8" × 13-3/8" annulus. The expansion created a restriction in the B-annulus, which prevented cement slumping and enabled placement of a continuous ~100 m cement plug across both annuli. Following expansion, the 825 Gator perforator created effective communication with the annulus, and circulation confirmed the well was static and free of hydrocarbons. A total of 250 bbl of seawater was circulated, ensuring the annulus was clean prior to cement placement.

Cementing operations achieved a balanced plug from 2192 ft (973 m) to 2864 ft (873 m) MD. The cement plug was subsequently tagged at the planned depth, and pressure testing to 1,250 psi for 15 minutes confirmed mechanical integrity and sealing capability. The plug held under applied load, verifying both placement and performance (see [figure 17](#) for the final well abandonment schematic).

The successful outcome demonstrated that the MPX provided the necessary structural support to prevent cement slumping, which had been a key concern for both the operator and the regulator. The regulator accepted the barrier as compliant with permanent abandonment standards. This represented a significant achievement, as the solution eliminated the need to cut and pull 9-5/8" casing, reducing operational complexity, rig time, and exposure to risk.

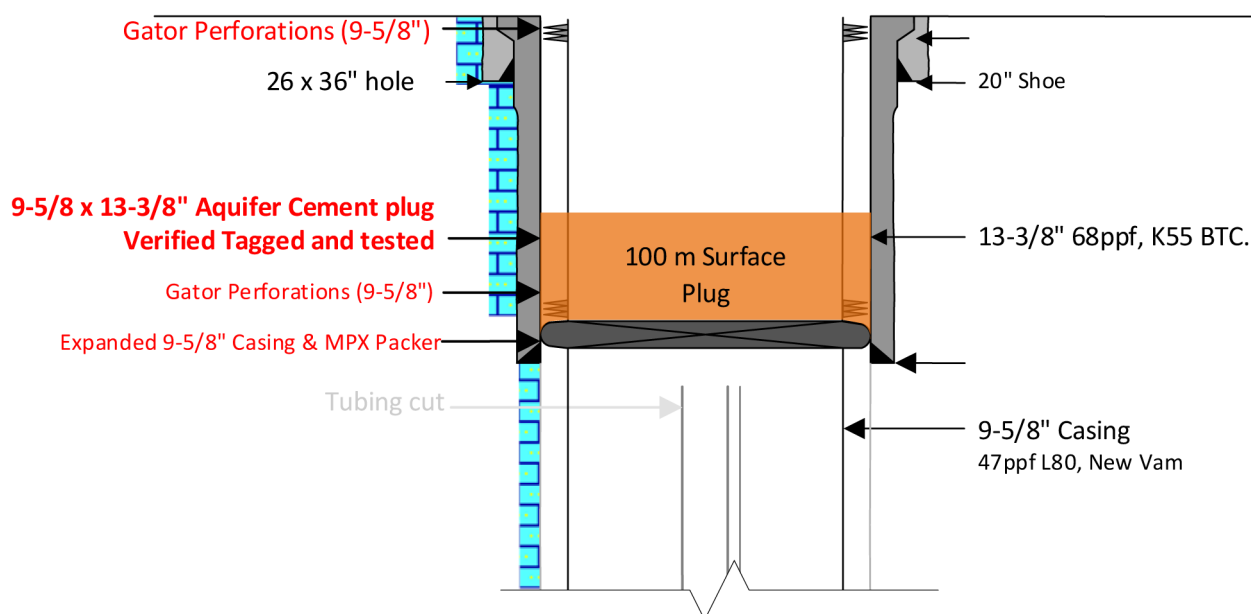


Figure 17—Final Well Abandonment

The results of this case confirm that the combined use of the hydro-mechanical expansion packer and mechanical perforator offers a reliable, verifiable, and regulator-approved method for achieving annular isolation in subsea well abandonments. The approach provides a repeatable solution for future projects where cement slumping, weak formations, or verification challenges make conventional methods insufficient.

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Nomenclature

AEL	Amplitude Energy Limited
ASIX	Mechanical Set Packer
BH	Big Hole
BHA	Bottom Hole Assembly
CSG	Casing
FPSO	Floating Production Storage and offloading vessel
Gator	Hydro-Mechanical Perforating Tool
ID	Internal Diameter
LES	Lee Energy Systems
LOT	Leak Off Test
MPX	Hydro-Mechanical Expansion Packer
NOPSEMA	National Offshore Petroleum Safety and Environmental Management Authority
NPP	Non Production Phase
OD	Outer Diameter
P&A	Plug and Abandonment
ROAM	Rotating Operator-Activated Annular Module
SGS	Static Gel Strength
ULS	Unloader Valve

VCV Velocity Control Valve

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